Phase information and vortex stretching

Mathematical review brief prepared for Rafael Lemus — 17 October 2026

Purpose and present status

The research question is whether a compact set of phase-sensitive observables improves reduced prediction of three-dimensional periodic Navier–Stokes dynamics. The existing phase-harmonic selection note provides a formal Taylor-series result for a restricted initial family. The experiments below test two stronger closure proposals. They do not establish global regularity or singularity formation. No claim of literature priority is made.

Use normalized averages on $[0, 2\pi]^3$ and define

$$E = \tfrac{1}{2}\langle |u|^2 \rangle, \quad Z = \tfrac{1}{2}\langle |\omega|^2 \rangle, \quad D = \langle |\nabla \omega|^2 \rangle, \quad S = \langle \omega_i \omega_j \partial_j u_i \rangle.$$

For smooth incompressible solutions, $Z' = S - \nu D$.

Test one An exact obstruction to a proposed power law

Consider the smooth divergence-free mean-zero fields

$$u_{A,k} = A\big(\sin ky, \sin kz, \cos kx - \cos k(x+y)\big), \qquad A > 0, \quad k \in \mathbb{N}.$$

Their initial diagnostics are exactly

$$E = A^2, \quad Z = \tfrac{5}{4}A^2k^2, \quad D = \tfrac{7}{2}A^2k^4, \quad S = \tfrac{1}{4}A^3k^3.$$

It follows that

$$\frac{S}{Z^{3/4}D^{1/5}} = \frac{A^{11/10}k^{7/10}}{4(5/4)^{3/4}(7/2)^{1/5}}.$$

Thus $S \le CZ^{3/4}D^{1/5}$ is false as a universal instantaneous inequality with finite C depending only on energy, fixed viscosity and domain size: set $A = 1$ and let integer k grow. A bandwidth-dependent constant is a different claim. A fit along selected trajectories is not contradicted by this argument, but cannot be promoted to this universal bound.

The numerical verifier evaluated 32 fields with $A \in \{1/2, 1, 2, 4\}$ and $k \in \{1, 2, 4, 6\}$ on grids 48^3 and 64^3. Diagnostics matched the exact expressions to maximum relative error 1.22×10^{-15}. The analytic scaling, rather than these finitely many samples, proves the obstruction. These are static diagnostic checks, not time integrations. At fixed positive viscosity, sufficiently high-k members are initially dominated by dissipation; this test does not produce blowup.

Test two Adding total stretching does not automatically close the dynamics

The pilot uses nine fixed wave vectors and transverse polarizations. Changing phases leaves every Fourier vector magnitude, and hence E, Z, D, unchanged. Viscosity is $1/28$ and the underlying equations are unchanged.

Parity construction. Take cosine-only fields, with all phases zero, then reverse the sign of the mode $(1, 1, 0)$ while retaining the other signs. Both fields obey $u(-x) = u(x)$. Therefore vorticity and the velocity gradient are odd, and the spatial average of their cubic stretching contraction is exactly zero. Both constructions have the same E, Z, D, S. Their common numerical diagnostics are approximately $0.5925, 1.1825, 6.835, 0$.

Centered directional differentiation of S along the discretized Navier–Stokes vector field gives, respectively,

$$\dot{S}_A(0) \simeq 0.6775039385, \qquad \dot{S}_B(0) \simeq 0.5426627067.$$

The observed gap is approximately 0.1348412318. The matching diagnostic mechanism follows analytically from parity and fixed modal magnitudes; these derivative values remain floating-point evidence, not interval-certified or exact rational evaluations.

Separate nonzero-stretching pilot. A phase root search finds two fields with matching E, Z, D and $S \simeq 0.0332666341$ to floating precision. The observed initial stretching rates are approximately 0.6949514716 and 0.5327836683. Spatial grids $24^3, 32^3$ and finite-difference increments were checked. RK4 integrations to $t = 0.1$, at time steps 0.002 and 0.001, give

$$S_A(0.1) \simeq 0.0993264842, \qquad S_B(0.1) \simeq 0.0843166812,$$

and $Z_A(0.1) \simeq 1.1648628382$, $Z_B(0.1) \simeq 1.1641267369$. The nonzero pair is matched numerically, not by a certified exact root. These are short-time convergence checks, not continuum error bounds or long-time regularity evidence. Full initial configurations, Fourier fields, scripts, and numerical outputs accompany the brief.

A focused question for mathematical review

Can a selected set of translation-invariant Fourier-triad observables predict, or control the error in predicting, vortex stretching and its evolution with bounds that remain controlled as spectral cutoff increases?

A practical benchmark should compare summaries (E, Z, D), summaries augmented by S, and summaries augmented by selected triad interactions. The targets should include future S, future D, and short-horizon Z on held-out initial fields, phases, amplitudes, spatial scales and viscosities. Using S to evaluate the identity $Z' = S - \nu D$ is not a forecasting result. Approximate prediction can be useful without an exact finite-dimensional closure; these must be evaluated separately.

For a regularity argument, a missing step is a uniform a priori estimate controlling the unresolved stretching over time, not merely an improved regression exponent. For a rigorous finite-time numerical claim, one needs validated residual and error bounds. For the parity pair, a modest formal next step is an exact Fourier calculation of the unequal stretching derivatives.

Materials needed to audit the separate strip-width chart

The newly supplied `hou luo.py` is a one-dimensional Hou–Luo model and measures a gradient-based width. It is not the generating code for the reported three-dimensional Fourier analyticity-strip chart. Audit of that chart requires its actual solver, initial data, raw spectra, fit windows, resolution criteria and stopping records. A fitted strip width reaching zero or a grid becoming underresolved is not itself a blowup proof.

Context and reproducibility

The accompanying package includes unchanged supplied periodic solver code, new test drivers, numerical outputs and a proposed benchmark protocol. See each experiment's README for commands and dependencies. The tests used supplied solver machinery; they are not an independent reimplementation of Navier–Stokes.

Relevant primary sources for an external review are the official Clay problem statement, Kang, Yun and Protas on finite-time enstrophy amplification, and Schumacher, Eckhardt and Doering on extreme vorticity growth. These studies provide context, not validation of the present experiments. Chen and Hou address smooth-data axisymmetric Euler blowup with a boundary; that is distinct from periodic viscous Navier–Stokes.